Experimental Reproduction and Analysis of *Transformers for Supervised Online Continual Learning*

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1 Introduction

Continual learning is an appealing research direction because it matches the way data arrives in real applications. Instead of training once on a fixed dataset, a model must keep predicting, updating, and adapting while the data distribution changes over time. This setting is difficult because models often suffer from *catastrophic forgetting*: when a new task arrives, previously learned knowledge is quickly overwritten.

In recent years, Transformer models have shown strong sequence modeling ability and remarkable *in context learning*, meaning that they can infer useful structure from recent context rather than relying only on previously optimized weights. From this perspective, Bornschein and colleagues ask a natural question: can the strengths of Transformers be used to solve *supervised online continual learning*? That question is studied directly in *Transformers for Supervised Online Continual Learning* [1], available at arXiv:2403.01554. On the CLOC benchmark, which exhibits natural temporal distribution shift, the paper shows that the answer is yes. When *in context learning*, *in weight learning*, and *replay streams* are combined, Transformer based models achieve a large improvement over earlier approaches.

This report summarizes the main ideas of the paper and presents a separate experimental reproduction on CIFAR 100 under the *piecewise stationary* protocol suggested by the authors. The experiments use the CIFAR 100 training split, normalize images with the standard mean and standard deviation of the dataset, and build an online stream with 100 consecutive tasks. In each task, the model samples 10 classes and then draws 5,000 examples with replacement from those classes. This yields a stream of 500,000 examples, processed in chunks of 50 examples per optimization step. The results show that the central message of the paper still holds on CIFAR 100: memory based methods such as Experience Replay and Transformer variants greatly outperform Online SGD, and the Transformer models also show clear signs of *meta learning* and *forward transfer*.

2 Summary of the Paper

2.1 Problem Setting and Main Proposal

Transformers for Supervised Online Continual Learning studies the online continual learning setting, where a model observes data sequentially through time and the data distribution changes continuously. Because the stream is *non stationary*, the model must both preserve previously acquired knowledge and adapt quickly to new changes. The objective used by the paper is the *cumulative next step prediction loss*: at each time step t, the model should predict y_t as accurately as possible given the full observation history up to that point.

The paper argues that the main limitation of earlier methods is that they rely almost entirely on storing information in model parameters through SGD. Approaches such as Experience Replay, ACM based on *kNN*, and Kalman Filter variants can help, but they do not fully exploit *in context learning*, which is a major strength of Transformers. As a result, even the best previous results on CLOC reached only about 38 percent. Figure 1 makes this gap clear: older baselines remain in the 20 percent to 38 percent range, whereas Pi Transformer reaches about 59 percent with supervised ResNet features, 67 percent when trained from scratch, and about 70 percent when combined with MAE ViT L.

Method	Pretrained	Finetuning	Avg. Acc.
ER	✓	✓	20%
ACM	✓	-	26%
Replay Streams	✓	-	≈ 38%
Replay Streams	-	✓	≈ 37%
Kalman Filter	✓	-	30%
Kalman Filter	-	✓	37%
Ours (no-image)	-	-	18%
Ours (no-attention)	✓	-	19%
Ours (Superv. ResNet)	✓	-	**59%**
Ours (MAE ViT-L)	✓	-	**70%**
Ours	-	✓	**67%**

Figure 1: Main results on the CLOC benchmark. The table highlights the large gap between Pi Transformer and earlier online continual learning methods.

This improvement does not come only from changing the feature extractor. It also comes from a unified design that combines two complementary mechanisms. The first is *in context learning*: the Transformer conditions directly on the most recent C observations through a KV cache ring buffer, with context size usually around 512 to 1024 tokens. This mechanism allows rapid adaptation when the distribution shifts suddenly. The second is *in weight learning*: the model is still updated online with SGD in the style of Transformer XL, with forward and backward passes performed on chunks of length $S \approx 256$. This component allows the model to accumulate long term knowledge and

improve steadily across tens of millions of observations.

The experimental narrative becomes more convincing when the full learning curves on CLOC are examined in Figure 2. The strongest Pi Transformer variants not only reach higher final accuracy, they pull ahead early and keep that advantage throughout online training. By contrast, the ablations such as "no image" and especially "no attention" drop sharply and almost return to the level of earlier baselines. This shows that contextual attention is not a minor detail. It is a central component of the model's fast adaptation ability.

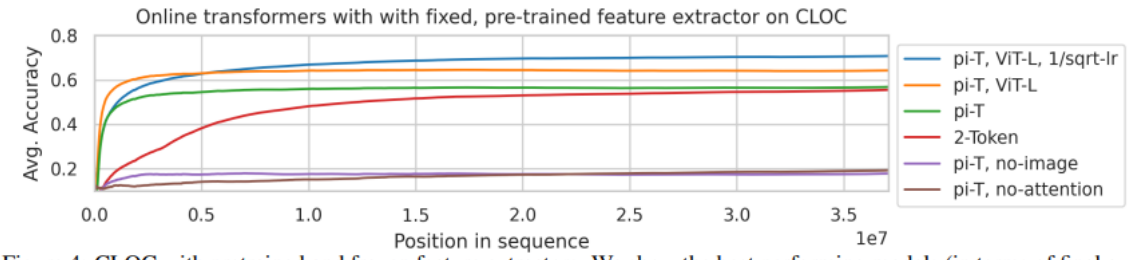

Figure 4: CLOC with pretrained and frozen feature extractors. We show the best performing models (in terms of final avg. accuracy) from the hyper-parameter cube in E.4 We also show pi-Transformer ablations: either without input features x_t, or without attention ($C = 0$).

Figure 2: Learning curves on CLOC for the main models and several ablations. The performance gap is maintained throughout training, not only at the final point.

The paper also introduces *Replay Streams*, where E parallel streams revisit older data in temporal order and are occasionally reset to earlier positions in the sequence. This design approximates multi epoch training without requiring a large replay buffer in memory. The model can therefore exploit short term contextual signal while also reinforcing long term knowledge from previously observed data. The two plots in Figure 3 illustrate this point directly: increasing the attention window improves average accuracy, and increasing the replay factor raises performance further while making training more stable at lower learning rates.

Figure 5: Stopping gradient updates at various positions to investigate the performance of *in-context* conditioning alone. Here for the pi-Transformer on CLOC with a fixed, pre-trained ResNet-50 feature extractor. Gradient based online-learning is most important at the beginning of the sequence, however keeps on contributing even after 10M examples.

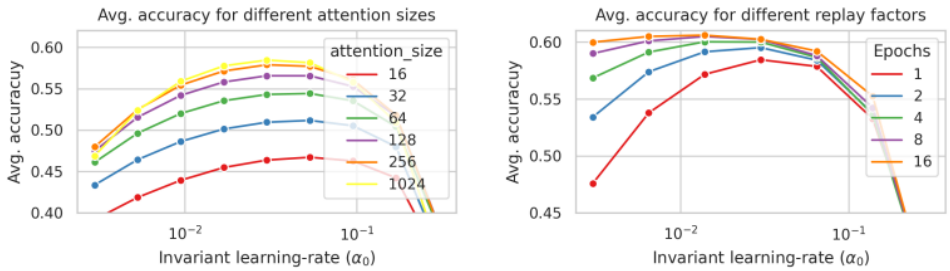

Figure 3: Effect of the attention window and the replay factor. Both components contribute directly to the performance of Pi Transformer.

Figure 4 provides a closer view of optimization. It shows that the number of replay streams must be

chosen in a way that matches the learning rate. If replay is too weak, the model barely benefits from past data. When replay is strong enough, accuracy rises substantially and remains stable across a wider range of learning rates. In other words, replay does more than remind the model of old data. It is the mechanism that lets online learning behave like multi epoch optimization while preserving a small memory footprint.

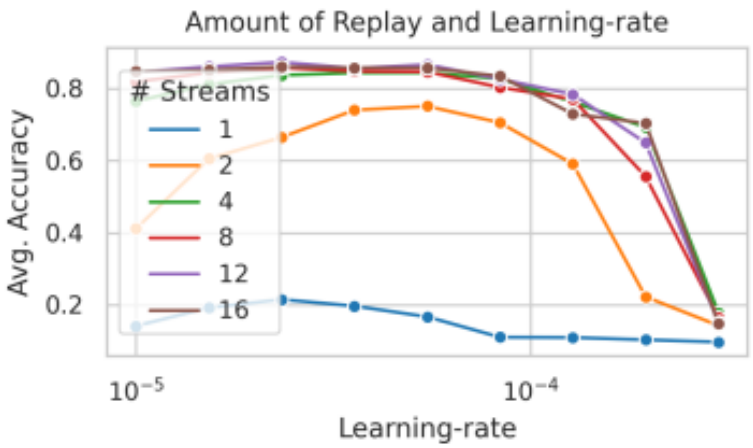

Figure 4: Joint effect of the number of replay streams and the learning rate on average accuracy.

The main architectural contribution of the paper is *Pi Transformer*, or *Privileged Information Transformer*. In this architecture, the label y_t enters the keys and values as privileged information, but the model is still not allowed to see y_t when it must predict that same label. Compared with a standard two token architecture, this design uses supervised signal more efficiently and improves convergence speed, training stability, and computational efficiency.

The paper also shows that conditioning on context alone is not enough. If gradient updates stop too early, performance drops substantially. This means that *in context learning* is powerful for fast adaptation, but it cannot fully replace *in weight learning*. Figure 5 shows that parameter updates matter most in the early phase, yet they continue to contribute even after very long training. This is key evidence for the paper's central message: the large performance gain appears only when contextual attention, replay, and parameter learning are combined in a single system.

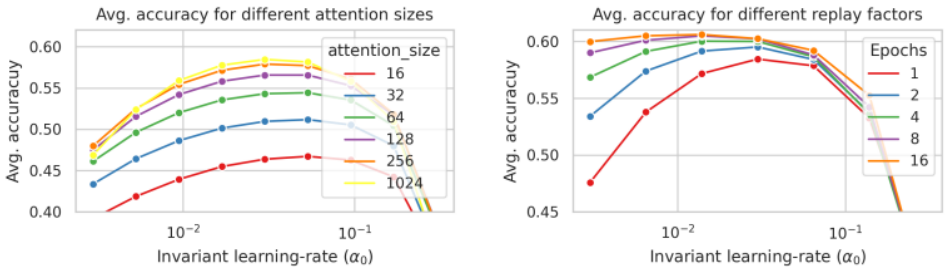

Figure 6: Pi-Transformer on CLOC with a fixed, pre-trained ResNet-50 feature extractor: **Left)** Increasing the attention window up to about 512 examples improves the results. **Right)** Performing more replay gradient steps improves performance and makes the model more robust to lower learning-rates.

Figure 5: Analysis of stopping gradient updates at different times. *In weight learning* continues to contribute for a very long period.

In short, the main claim of the paper is that a strong online continual learning system should not rely only on packing all experience into model weights. Instead, it should combine fast adaptation in context with long term accumulation in parameters. The interaction among *in context learning*, *in weight learning*, replay streams, and the Pi Transformer architecture is what produces the large performance jump reported by the paper.

3 Experiments on CIFAR 100 and Analysis

3.1 Experimental Setup

Dataset and stream construction. The experiments evaluate on CIFAR 100, following the *piecewise stationary* protocol described in Appendix D of Bornschein and colleagues [1]. CIFAR 100 contains 50,000 training images across 100 classes, with 500 images per class. In the online continual learning setting, no held out test set is used. Every prediction made during training is also a test prediction. This is the *prequential evaluation* protocol: the model predicts before it learns.

The stream is constructed as follows. The setup uses 100 sequential tasks. For each task, 10 distinct classes are drawn from the 100 classes of CIFAR 100, and then 5,000 examples are sampled with replacement from those 10 classes. The full process creates a stream of 500,000 examples. At every task boundary, the 10 selected classes receive fresh local labels from 0 to 9 independently. As a result, the same image can receive different labels in different tasks, and there is no fixed mapping from visual content to label across the whole stream.

This design is deliberate. Because labels are remapped from task to task, visual features alone cannot determine the current label assignment. The model must instead read recent context, namely the recent (image, label) pairs observed in the current task, in order to infer the active mapping. This is exactly the regime where the Transformer's *in context learning* mechanism is expected to matter.

The stream is consumed in chunks of 50 examples per gradient step, for a total of 10,000 training

steps. At each step, the model predicts labels for the current chunk before seeing the true labels, records accuracy, and then updates its weights. Each example is seen exactly once in the main stream. There is no global shuffling, no multiple epochs, and no explicit task identity signal.

Feature extractor. The experiments use the VGG++ architecture described in the paper, a VGG inspired convolutional network that replaces BatchNorm with GroupNorm and uses 10 percent dropout. VGG++ is trained jointly with the Transformer from random initialization and receives gradients at every step alongside the temporal model. This matches the intended setting of the paper, where the feature extractor and the temporal model adapt together online.

GroupNorm is used instead of BatchNorm because BatchNorm depends on statistics computed across the batch. With chunk size 50, those statistics are unstable and can leak temporal information across examples in the same chunk, which violates the online protocol. GroupNorm normalizes each sample independently and is therefore more robust for small batch sizes.

Methods compared. Four methods are trained on the same stream. *Online SGD* is the lower bound: it uses only a linear classification head on top of VGG++ features and updates that head with AdamW, without any memory mechanism. *Experience Replay* maintains a reservoir buffer of past feature and label pairs and replays stored examples alongside the current chunk. *Pi Transformer* is the architecture proposed by the paper, using a 4 block Transformer with Multi Query Attention. Past labels are injected into the key and value matrices of each attention layer, and a zero diagonal causal mask prevents the model from reading its own target label. *Two Token Transformer* is the comparison architecture in which each example becomes two tokens, one for the image feature and one for the label, with the current label token masked. It uses the same 4 block structure and the same replay configuration as Pi Transformer.

Hyperparameters. All Transformer methods use 4 blocks, model width 256, chunk size $S = 50$, constant learning rate 10^{-4}, the AdamW optimizer, 20 replay streams, and context window $C = 512$, except in the ablations where one component is varied.

3.2 Main Results

Table 1 summarizes the main metrics. The gap between Online SGD and the three memory based methods is very large. Online SGD reaches average accuracy 0.3291, while Pi Transformer reaches 0.7426, Experience Replay reaches 0.7580, and Two Token Transformer reaches 0.8346. If only the last 20 tasks are considered, the gap becomes even clearer: Online SGD reaches 0.4197, while Pi Transformer reaches 0.8798, Experience Replay reaches 0.9589, and Two Token Transformer reaches 0.9696.

Table 1: Comparison of methods on CIFAR 100 *piecewise stationary*.

Method	**Avg. Acc.**	**Last 20**	**BWT**	**FWT**
Online SGD	0.3291	0.4197	−0.2725	−0.0780
Experience Replay	0.7580	0.9589	0.2393	0.2521
Pi Transformer	0.7426	0.8798	0.1874	0.1253
Two Token Transformer	0.8346	0.9696	0.1549	0.1873

Figure 6 makes this result visually clear. The gray dashed curve of Online SGD remains at a low level across the 500,000 example stream and exhibits the familiar sawtooth pattern: each task switch causes accuracy to collapse, and only partial recovery occurs before the next task begins. This is classic catastrophic forgetting. By contrast, the three memory based methods improve steadily over time. Task boundary drops remain visible, but they become shallower and recovery becomes faster as more tasks are experienced.

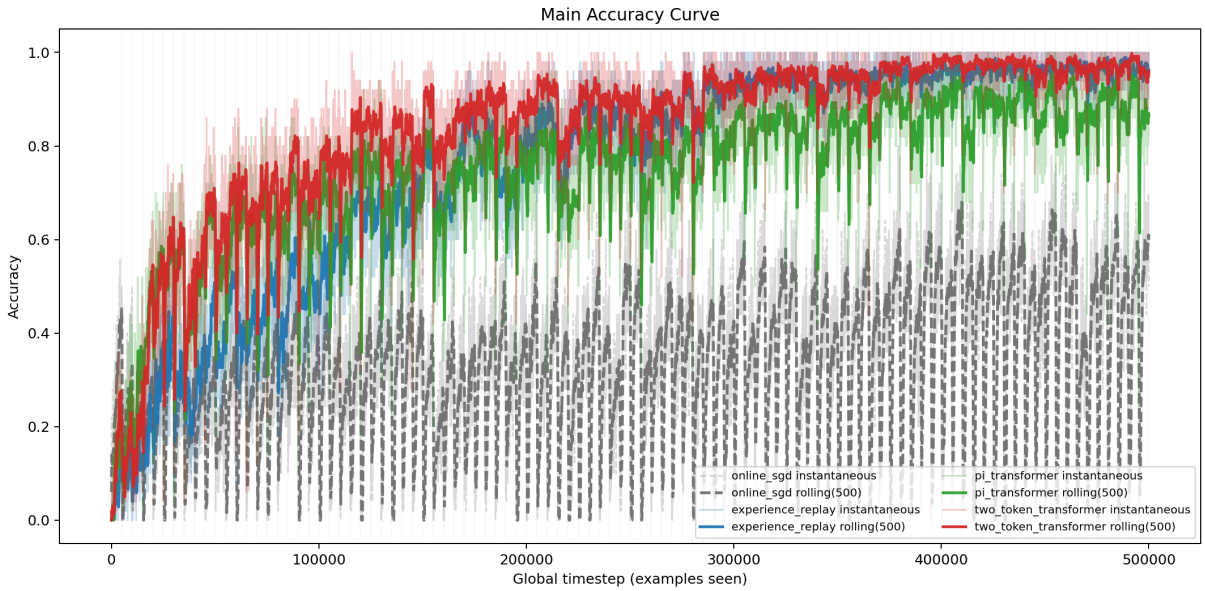

Figure 6: Accuracy over the full 500,000 example stream. Online SGD keeps the characteristic sawtooth pattern of catastrophic forgetting, while Experience Replay and the Transformer variants improve steadily with time.

Quantitatively, Pi Transformer exceeds Online SGD by about 41 percentage points in average accuracy and about 46 percentage points on the last 20 tasks. This confirms the central claim of the paper: combining *in context learning* with *in weight learning* through replay is far more effective than pure gradient based adaptation without memory.

3.3 Catastrophic Forgetting Is Largely Solved

Backward Transfer, or BWT, measures whether performance on old tasks improves or degrades over time. Negative BWT means the model forgets old knowledge. Positive BWT means the model preserves or even improves old knowledge as training continues. Figure 7 shows that Online SGD has BWT equal to -0.2725, meaning that after the full training process its accuracy on old tasks is on average more than 27 percentage points below the level reached immediately after those tasks were first learned.

By contrast, all three memory based methods have positive BWT. Experience Replay reaches 0.2393, Pi Transformer reaches 0.1874, and Two Token Transformer reaches 0.1549. This indicates that past knowledge is not destroyed. It continues to be reinforced during online training. On this CIFAR 100

piecewise stationary benchmark, catastrophic forgetting is therefore largely resolved for the replay based and Transformer based methods.

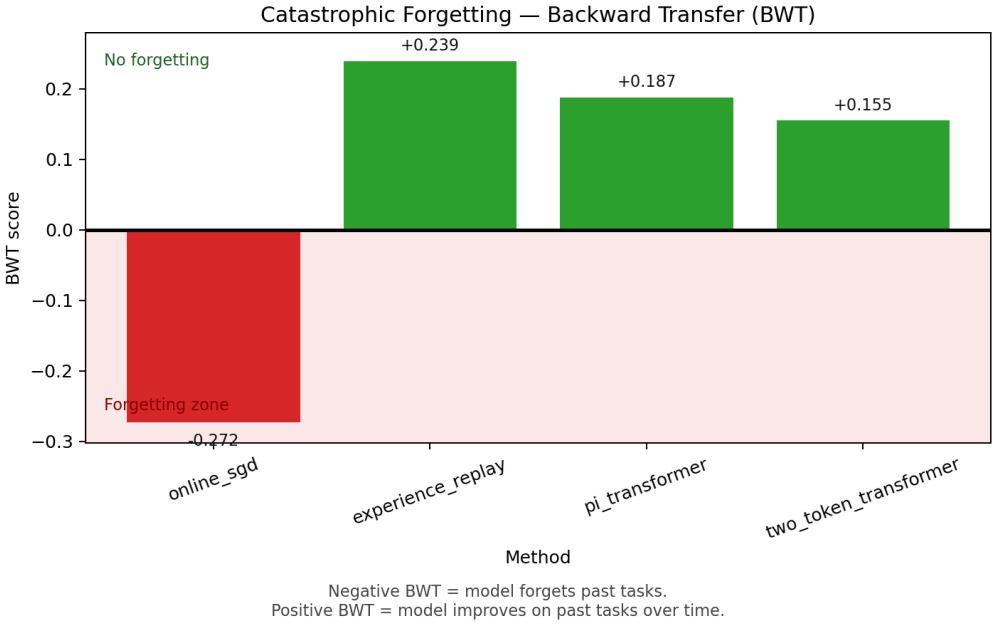

Figure 7: Backward Transfer for the four methods. Online SGD remains in the forgetting region, while the three memory based methods all have positive BWT.

3.4 The Model Becomes a Better Learner Over Time

Another important question is whether the model becomes better at learning new tasks as it accumulates experience. Figure 8 answers this clearly. For Online SGD, mean accuracy within each task fluctuates around 25 percent to 45 percent from beginning to end, with almost no upward trend. This suggests that the method does not acquire a general adaptation strategy.

The memory based methods show the opposite pattern. Pi Transformer starts near 20 percent and rises toward about 88 percent on late tasks. Two Token Transformer rises even faster and reaches about 95 percent relatively early. Experience Replay shows a similar upward trend. This is evidence of *forward transfer*: knowledge from previous tasks makes the model systematically better at adapting to new ones.

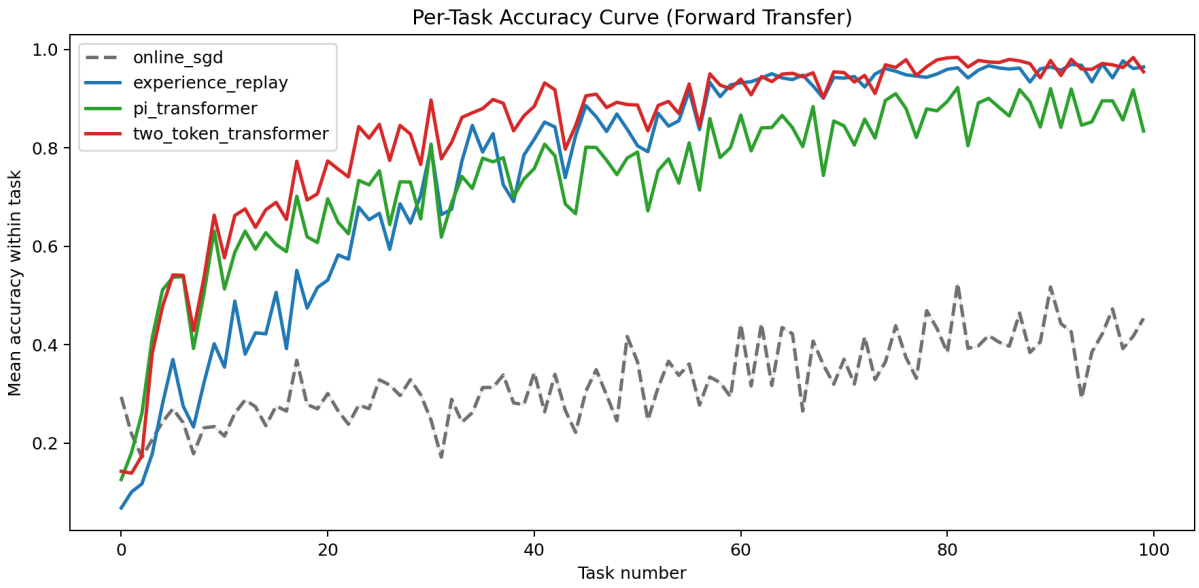

Figure 8: Mean accuracy within each task as a function of task index. Memory based methods show clear *forward transfer*, while Online SGD changes very little.

This interpretation is supported by the detailed within task recovery curves in Figure 9. Early in training, accuracy remains near zero until the first chunk has been seen and then rises only slowly. Late in training, recovery after a task boundary is much faster. The model begins to infer the current label mapping from recent context much more efficiently.

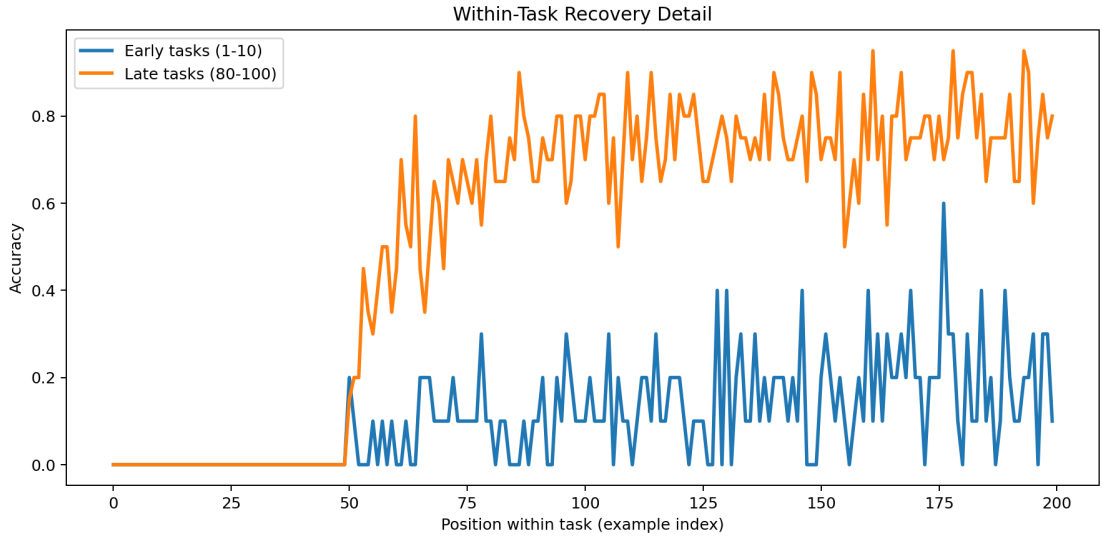

Figure 9: Within task recovery curves. Adaptation after a task boundary becomes substantially faster later in training.

Figure 10 summarizes the same trend more compactly. Instantaneous accuracy is higher on late

tasks than on early tasks, confirming that the model is not only retaining knowledge but also becoming a better learner over time.

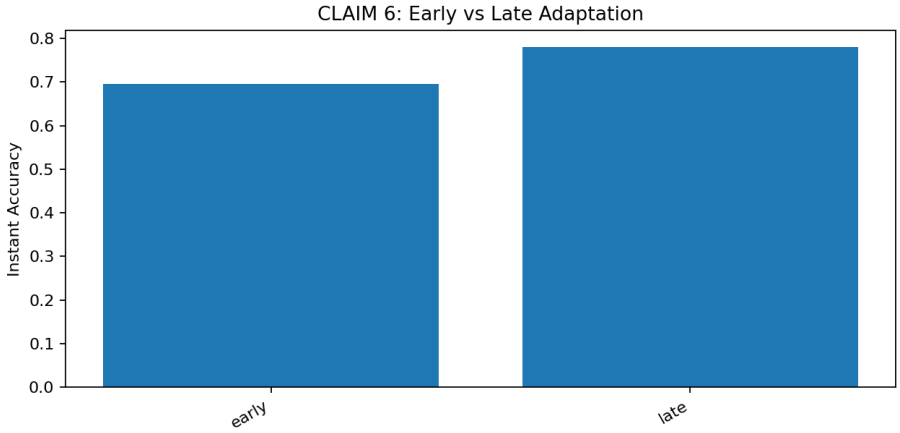

Figure 10: Comparison between early tasks and late tasks. Late tasks are solved more effectively, showing clear forward transfer.

3.5 Replay Streams Are the Main Driver

The ablation over the number of replay streams in Figure 11 shows that replay is the crucial component. When $E = 1$, which is essentially no replay, rolling accuracy is about 0.297. Increasing replay to $E = 2$ raises accuracy to about 0.391, $E = 4$ reaches about 0.565, $E = 8$ reaches about 0.710, and $E = 20$ reaches about 0.877. The staircase shape of the curve is striking: every additional replay stream helps, and there is no reversal.

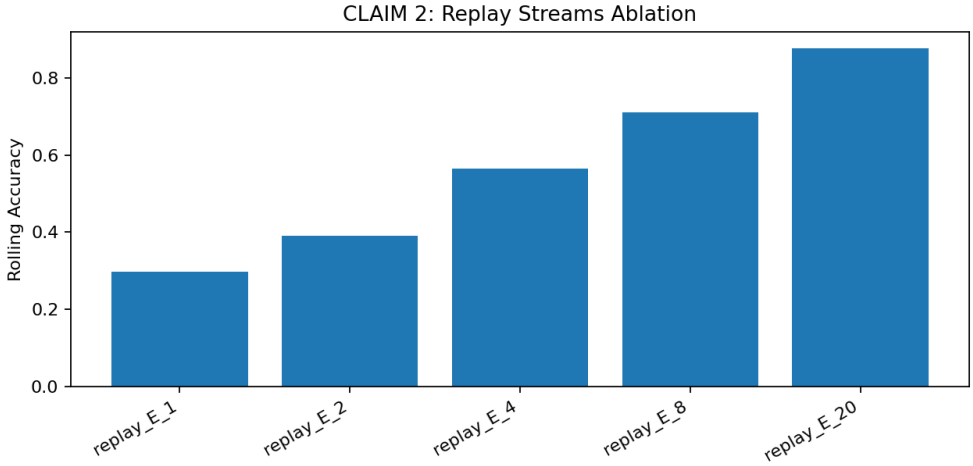

Figure 11: Ablation over the number of replay streams. Increasing E from 1 to 20 yields a strong and nearly monotonic gain.

Mechanistically, this can be understood as follows. Without replay, the model optimizes only at the current position of the stream, so weights quickly specialize to the current task and overwrite previous knowledge. With replay, the model must perform well both on the present position and on randomly sampled past positions. That pressure encourages a more general strategy over the whole stream, namely reading the label mapping from recent context rather than trying to memorize every task in the weights.

3.6 Context Window Ablation

Figure 12 shows a particularly interesting result when the context window size C is varied. Unlike the original paper on CLOC, where larger windows usually improve performance, the CIFAR 100 setting of this report favors small windows. Values such as $C = 1$, $C = 16$, and $C = 64$ achieve the best recovery accuracy in the first 100 examples of each task, at about 0.3553, 0.3680, and 0.3702 respectively. When the window is increased to $C = 128$, $C = 256$, and $C = 512$, recovery accuracy drops sharply to 0.2861, 0.2462, and 0.2164.

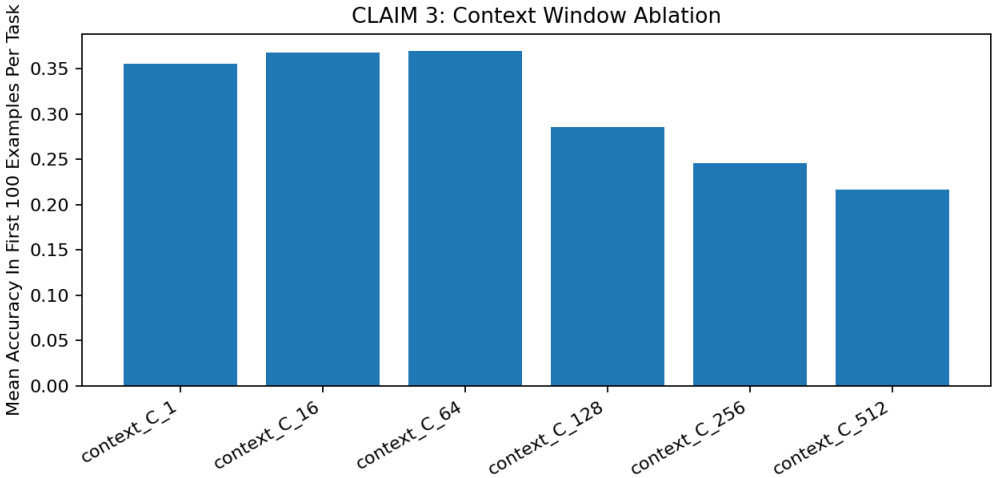

Figure 12: Ablation over context window size. In the CIFAR 100 *piecewise stationary* setting, larger windows do not improve rapid recovery after a task boundary.

One plausible interpretation is *context poisoning*. At each task boundary, the context memory is filled with examples from the previous task under a label mapping that is no longer correct. If the window is too large, the model faces a large amount of highly confident but wrong signal exactly when it needs to adapt quickly. In an environment with abrupt task boundaries and task specific label remapping such as CIFAR 100 *piecewise stationary*, a long context window can therefore become a disadvantage. This appears to be an important boundary condition that the original paper did not test directly on CLOC, where the distribution changes more smoothly and past context remains useful.

3.7 Both Mechanisms Are Necessary

Figure 13 shows the freeze ablation. It compares the full method, freezing Transformer weights at step 5,000 while keeping context active, and removing the context window while leaving SGD active. The recovery metric shows that removing the context window can look favorable in the very first 100 examples of a task, while freezing weights hurts much more strongly.

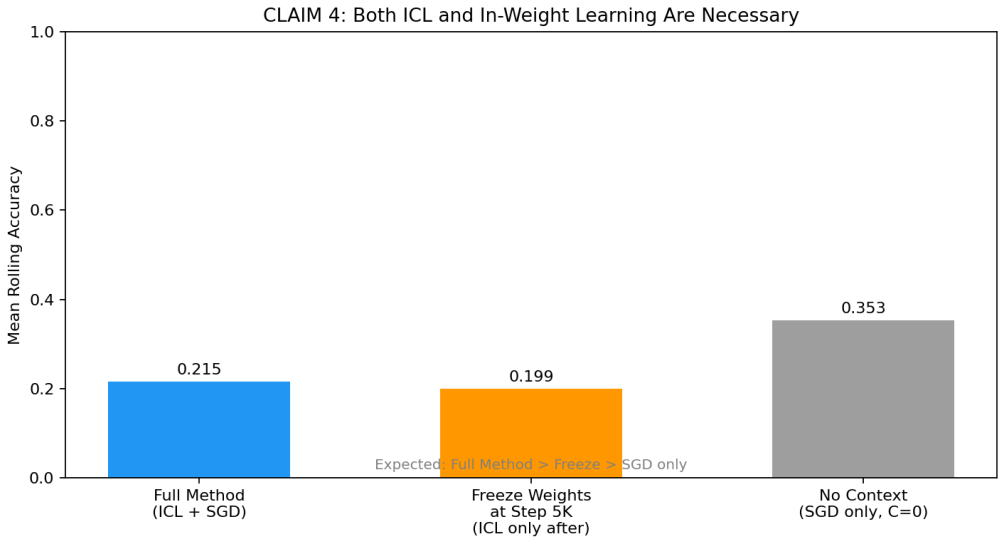

Figure 13: Freeze ablation. Both contextual memory and ongoing weight updates matter, even though their contribution depends on the metric being measured.

The seemingly surprising result, where no context can recover faster right after a task switch, is consistent with the context poisoning argument above. Without context, the model avoids the misleading signal left by the previous task and relies only on its weights. However, the broader picture still favors the full method. The full system has positive backward transfer, while the pure SGD style baseline forgets old tasks badly. Fast local recovery and long term knowledge retention are different goals, and they depend on different mechanisms. Context supports one, and weight updates support the other.

3.8 Architecture Comparison

When the two Transformer architectures are compared directly, Figure 14 shows that Two Token Transformer outperforms Pi Transformer on this benchmark. Mean rolling accuracy for Pi Transformer is about 0.884, whereas Two Token Transformer reaches about 0.965. This differs somewhat from the emphasis of the original paper, where Pi Transformer is presented as especially stable and particularly useful when labels have rich structure, such as GPS coordinates on CLOC.

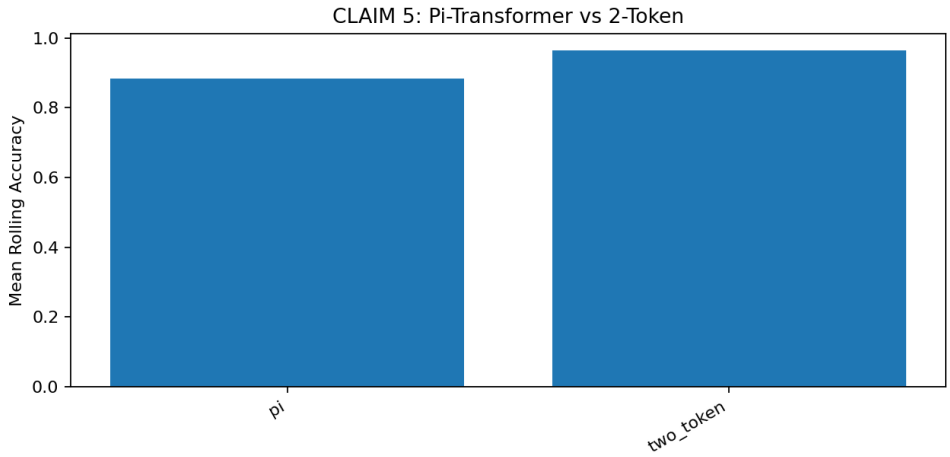

Figure 14: Comparison between Pi Transformer and Two Token Transformer on CIFAR 100 *piecewise stationary*. Two Token Transformer performs better in this setting.

One reasonable explanation is that the label spaces of the two benchmarks are very different. On CLOC, labels are structured and continuous, so the privileged information mechanism of Pi Transformer can provide a stronger advantage. On CIFAR 100, labels are just discrete integers from 0 to 9 within each task, so the Two Token design, where the label enters the sequence directly as a token, may be a simpler and more effective representation.

3.9 Effect of the Feature Extractor

To better understand the role of feature representation in online continual learning, this report compares two feature extractor configurations on the same CIFAR 100 *piecewise stationary* stream. The first configuration uses a ResNet 50 pretrained on ImageNet and keeps all feature weights frozen throughout training. The model receives precomputed features, and only the temporal head is updated. The second configuration uses VGG++, a VGG style convolutional network with GroupNorm and 10 percent dropout, trained jointly with the Transformer from random initialization. This is also the configuration described in Appendix D of the paper for the CIFAR 100 experiments.

Figure 15 shows the average accuracy across all 100 tasks for both configurations. The pattern is consistent and informative. For Online SGD, jointly trained VGG++ performs much better, around 0.33, than frozen pretrained ResNet 50, around 0.24. Although pretrained features are initially stronger, Online SGD has no memory mechanism that can compensate for the mismatch between ImageNet features and the task local 10 class prediction problem on CIFAR 100. When trained jointly, VGG++ can adapt together with the classifier head and gradually learn features aligned with the current label mapping in the stream.

For all memory based methods, however, pretrained features are slightly stronger than jointly trained VGG++. Experience Replay drops from about 0.82 to 0.76, Pi Transformer drops from about 0.77 to 0.74, and Two Token Transformer drops from about 0.86 to 0.83. These gaps are small, roughly 3 to 6 percentage points, and they reflect a clear cold start effect. When training begins from random VGG++ features, the first few thousand examples are processed with a weak representation. The

Transformer must therefore learn to use context while also waiting for the feature extractor to become discriminative enough. This slows adaptation in the early phase and lowers the mean across all 100 tasks.

The important point is that the ranking among methods remains unchanged across both feature settings. Two Token Transformer still exceeds Pi Transformer, Pi Transformer still exceeds Experience Replay, and all three remain far ahead of Online SGD. This stability suggests that the qualitative conclusions of the report are not artifacts of a particular feature extractor. Context poisoning at task boundaries, emergent meta learning, and the central role of replay all remain visible in both configurations.

3.10 Conclusion on the Feature Extractor

Figure 15: Average accuracy for frozen pretrained ResNet 50 features versus jointly trained VGG++ features on CIFAR 100 *piecewise stationary*.

This comparison supports two complementary conclusions. First, feature quality sets the performance ceiling for the entire system. Strong pretrained features give memory based methods a clear advantage, especially at the beginning of the stream when joint training has not yet produced a strong representation. Second, joint feature learning still matters. For methods without memory, the ability to adapt the representation to each task is the main source of improvement, and VGG++ does this much better than a frozen general purpose extractor.

With respect to reproducing the paper faithfully, the jointly trained VGG++ setting is the correct one. The paper states that the CIFAR 100 experiments use this configuration, and the results in this report confirm that it reproduces the intended qualitative behavior, including forward transfer, the need for replay, and the emergence of *in context learning*. The 3 to 6 percentage point advantage of pretrained features is therefore a practical cold start effect rather than a change in the scientific conclusion.

4 Experimental Conclusion

Taken together, the CIFAR 100 *piecewise stationary* results confirm the core claim of Bornschein and colleagues. First, memory based methods, especially Experience Replay and the Transformer variants, outperform Online SGD by a very large margin. Second, the Transformer models do not simply retain old knowledge. They also show clear signs of *meta learning*: as the number of tasks grows, they adapt more quickly to new tasks. Third, replay streams play a central role by forcing the model to learn a strategy that generalizes across the entire stream.

At the same time, the experiments in this report identify an important condition for the advantage of Transformers. *In context learning* works best when past context remains informative for the present. On CLOC, where distribution shift is gradual and relatively smooth, that condition holds and Transformer models are clearly superior. On CIFAR 100 in the *piecewise stationary* regime, abrupt task boundaries and local label remapping can make past context actively misleading. As a result, a large context window is no longer always helpful. This suggests an important practical recommendation: when a deployment setting has abrupt phase changes, smaller context windows or explicit cache clearing mechanisms may be preferable.

The comparison between frozen pretrained ResNet 50 features and jointly trained VGG++ adds another useful layer of interpretation. Pretrained features help memory based methods start from a stronger representation and achieve slightly higher mean accuracy, while joint training is especially valuable for a method without memory such as Online SGD. Therefore, the choice of feature extractor mainly affects early adaptation speed and the short term performance ceiling. It does not change the ranking among methods or the main scientific message of the paper.

In short, the central contribution of the paper remains valid: combining Transformer *in context learning* with replay based *in weight learning* produces a much stronger online continual learning system than pure gradient optimization. The CIFAR 100 study also clarifies when this advantage is strongest and when it must be adjusted to avoid *context poisoning*.

References

[1] Jörg Bornschein, Yazhe Li, and Amal Rannen-Triki. *Transformers for Supervised Online Continual Learning*. CoRR, abs/2403.01554, 2024. https://doi.org/10.48550/arXiv.2403.01554.